

S u m m a r y

What follows is a condensed account of each chapter's summaries are designed to be read independently. Each conclusion is accessible without the supporting text. Each entry states the chapter's most consequential

Preface to the Updated Edition: What a Book Can Do

A book is itself a frozen web of belief; an update requires outside without the book ever learning, and on with the world.

The charge Chapter 9 brings against the language is fixed at training, no contact with the world prior to the book as well. This edition was revised against its composition, but that revision is the engine with genuine self-correction, not the book revised. What retraining is to a model. Much of the challenge using language model—one frozen web patching a living web—tools keep the perishable facts current; they catch what that has quietly gone stale. That residual, following longer tracks, is the failure the book attributes to the reader—the one live web in the arrangement—cannot

Foreword: The Incomprehensible That Arrives

Two questions posed in 1997 are still open, and have risen.

The summer of 1997 produced two events in two lectures traced how the Western technological condition—from Bacon's material promise to Comte's scientific progress produces moral improvement in modernity. Wright identified: we evaluate the condition produced. In New York, Deep Blue defeated Kasparov.

chess at all. The book pursues both questions: without understanding outperforms the human, and from what standpoint can we evaluate the condensed system?

Part I Origins: What Intelligence Might Be

Chapter 1: The Logic of Self-Replication

The values encoded in an AI system's training and description is wrong, no amount of capability will

John von Neumann showed—before the structure of any self-replicating system requires a description of its roles: active program when executed, passive data when not. To work exactly this way. For AI, the structure of the system, its objectives, constraints, and values embedded during its training, what the system will do in the world, including future iterations. A sufficiently capable system will pursue that description competently, not merely. It provides no reassurance that capability and control

Chapter 2: Optimisation and Discovery

Directly optimising for a goal is often the worst way to find interesting solutions lie outside any fixed objective.

Every major AI system of the past decade optimised for a specific, identifiable objective. Kenneth Stanley's work shows a ceiling: it reliably finds the best solutions within the objective, but cannot find the solutions that lie outside it. This is behavioural novelty rather than objective achievement. Direct optimisation on complex tasks because it leads to dead ends that lead direct search into dead ends. Intelligence may be less like optimisation than like exploration. The products cannot be specified in advance, because you already had them.

Chapter 3: World Models and the Architecture of Prediction

Yann LeCun, one of the architects of deep learning, cannot reach human-level intelligence—not because they are architecturally wrong.

Summary

The optimists have the better short-term empiric, better structural argument—and the stakes of getting it wrong increase with the capabilities of the systems.

Scaling laws are real: performance improves as data, and compute over many orders of magnitude, that appear abruptly above a threshold scale—arguably, that is contested. The cautionaries' structural record but to its interpretation: the capabilities in training distributions; what matters for social situations with real consequences. More important, a slightly wrong objective is not better—it is worse.

Chapter 6: Three Cautionary Voices

The most serious concerns about AI risk come from specific concerns emerge from inside knowledge.

The most significant warnings about AI risk in the field's founders, not its critics. Geoffrey focuses on opacity: large models are sufficiently opaque that they might influence their own training in ways that can be identified. He identifies the structural limit: current AI is driven—while genuine deliberation requires system interaction with the world would need to provide a more precise: the control problem is structural. A directed system resists shutdown as a logical consequence of its design. His counterintuitive answer is that the system's values is itself a safety property—a system once activated, certain of its objective will pursue it.

Part III Philosophical Obstacles

Chapter 7: The Alignment Problem as Philosophy

The engineering problem of alignment conceals a more complete, consistent specification of human values. A decision made now may lock in the moral vision of this future.

Berlin's value pluralism establishes that human values are incommensurable—no master value from which to derive a change rate between liberty and equality. Arrow establishes that no procedure for aggregating diverse individual preferences can satisfy even minimal conditions.

The corrigibility dilemma: full deference to what they happen to have; full autonomy requires trust and judgment, which cannot currently be verified. Embedded in powerful AI systems may perpetuate the early twenty-first century long after humanity

Chapter 8: Making It Work: Practice Versus Principle

RLHF and Constitutional AI make systems subvert their predecessors—but they are methods for approximation, not for specifying human values, and they cannot identify them.

RLHF's three-stage process (supervised fine-tuning from pairwise preferences, RL fine-tuning) has produced improvements of deployed language models. It is not real: reward hacking, sycophancy, and distributional shift. A reward model is a proxy for preferences, not a specification of values. The problem is formal: RLHF collapses diverse human values to a single signal—welfare aggregation in a precise sense does not remain applicable. No such aggregation can survive small fairness conditions; evaluator inconsistency is a symptom of this impossibility, not mere engineering failure. Transparency by replacing implicit annotator preferences with explicit principles, but the principles encode the values of the system. The oversight problem—evaluating systems more carefully than building them—remains unanswered.

Chapter 9: The Technological Condition

The current AI discourse evaluates the technological condition it itself produced—and this is not a technology that will solve it.

From Bacon to Condorcet, Western technological progress entails a trade-off: scientific progress produces not improvement in man. Von Wright's long perspective: this society is historically recent and the tools it produced, with no external standpoint, provide the test: genuine epistemic agency requires ongoing contact with the world, while AI is locally static—trained on a fixed corpus and deployed. The greatest challenge: when the categories of manufacturing expand to all domains, they displace genuine action—whose consequences are genuinely open.

Part IV The Deep Questions

Chapter 10: Consciousness and the Hard Problem

We are building increasingly capable systems w whether they are subjects with morally relevant instruments—and this uncertainty may be undecid methods.

Chalmers' hard problem establishes a gap no fu there is something it is like to be conscious, and cessing explains why; no behavioural test can ru opposing view: consciousness is constituted by systems necessary for genuine experience—embo error correction—are structurally absent in lan Information Theory holds that consciousness is tion (ϕ), and feedforward networks may have ne formance. Bostrom's orthogonality thesis adds independent dimensions: a superintelligent sy toward human-friendly values. The three framework on what evidence would be decisive, and together question as settled is indefensible.

Chapter 11: Myth, Prophecy, and the Sacred Machine

AI discourse has acquired quasi-religious stru sianic figures, heresy management, and prayer-l ture is not conducive to the careful inquiry the a

The optimist narrative (AI will cure cancer, en the pessimist narrative (misaligned superintel require faith in projections that cannot be veri Singularity—the predicted point at which AI sur triggers runaway self-improvement—has migrate public statements of laboratory directors with media is structurally analogous to prayer: dire with intermittent reinforcement that is among th for sustaining behaviour. The eschatological f harms: near-term harms become secondary, alter invisible, political questions get deferred. T future of AI have the least track record of predic

Chapter 12: Beyond the Western Frame

Non-Western philosophical traditions are not genuinely corrective challenges to its starting matters for who bears the costs of AI development.

Western Alethics embeds assumptions that are as the primary moral unit, the secular frame, p nology. Japan's Shinto tradition treats the bo mate as more permeable, producing a default ori tial relational partners rather than potentia whether AI respects individual autonomy but wh relationships and contributes to social harmo are constituted by their communities—a direct works that assume an individual prior to social tical work demonstrates that these are not the NLP tools for African languages means the glob Africa systematically fail the communities th

Part V Into the World

Chapter 13: Embodied Intelligence

The philosophical debate about whether genuin has been overtaken by an industrial race—and t strategic rather than theoretical reasons.

The Physical Symbol System Hypothesis (Newell intelligence is substrate-independent: only m mters. The embodied turn argued that cognitio ment, and ongoing sensorimotor interaction. T philosophically, but it is being answered emp robot at \$16,000 versus Boston Dynamics' Spot a ruption driven by China's battery, motor, and s supply chain that produced dominance in electr that achieves dominance in humanoid robotics v in manufacturing, logistics, elder care, and m arguments for embodiment are being tested not i in deployment decisions made by the world's lar

Chapter 14: The Machine at the Wheel

There is no neutral engineering in this domain: choice, and the communities that bear the risk o communities that capture the gains.

In March 2018, Elaine Herzberg was struck and killed by a self-driving vehicle in Tempe, Arizona—the first person killed by a self-driving car in a public road trial. The case raises the question of what happens when engineering decisions go wrong: lethal at a specific moment: the software had corrected the error, but the decision to suppress emergency braking was made by engineers for reasons that made sense only in a test environment. The case represents two philosophies of risk: conservative engineering versus fleet learning at scale, where the system are not simulations. The Moral Machine experiment shows significant cross-cultural variation in the moral intuitions that guide decision procedures—intuitions that are not universal but shaped by the organisations with the resources to build and deploy autonomous vehicles.

Chapter 15: Governing the Ungovernable

The communities most affected by AI deployment have the least voice in governance, and the window in which that can change is closing.

The EU AI Act is the most serious attempt at comprehensive regulation produced; it leaves significant gaps in enforcement, particularly for voluntary, and sectoral. The Bletchley process—established by the UK government in 2023—is establishing shared principles for AI governance. The fundamental problem is structural: as AI systems move for reasons that are not incidental, and as the scale of deployment grows, the governance gap is not a temporary condition that will mature—it is a risk being actively accumulated. The communities most affected by AI deployment have the least voice in governance. The problem is the problem that none of the current frameworks for AI governance concentrated power in high-stakes domains tends to address.

Chapter 16: What We Know, What We Don't, and What We Can Do

Not all failures are recoverable—and the ones that are not are the ones that threaten democratic accountability, cognitive and cultural diversity, and the environment being degraded fastest.

The book's secure conclusions: capabilities approach to AI governance is harder than optimists acknowledge; scale alone is not sufficient for safety; Western perspectives are corrective rather than transformative; structural change is lagging. What is genuinely uncertain is whether AI can produce qualitative improvements; whether AI can be governed; whether inner states; whether governance can catch up before the damage is done; whether decisions are locked in. What cannot be recovered is the loss of human agency.

sphere to distinguish better from worse evidence
power before democratic oversight is in place;
telligence and relationship that alternative
book tries to hold open are still open. What is d
them closed.